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Descriptive proximity-based modeling for accurate and explainable battery state-of-charge estimation

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ABSTRACT

State-of-Charge (SoC) estimation of batteries is critical for electric vehicle health management. However, achieving high accuracy under heterogeneous operating conditions while preserving model transparency remains challenging. This paper proposes a novel descriptive proximity-based learning framework for accurate and explainable SoC estimation. Battery discharge data are mapped into a descriptive feature space, $\Phi(x)$={current, voltage, capacity, energy, cycle count, temperature}, where local proximity reflects similar operational states. An adaptive ϵ-proximity threshold is derived using robust Median Absolute Deviation (MAD) statistics. This threshold guides a density-based clustering algorithm (DBSCAN) to identify intrinsic regimes without manual tuning.

We then define fuzzy proximity relations between clusters and rigorously prove that they satisfy the reflexivity, symmetry, and transitivity-like axioms of proximity. A gradient-boosted tree model (LightGBM) is trained using the cluster-informed features, yielding near-perfect SoC predictions (e.g., cross-validation RMSE $\approx$ 0.13 and $R^2 \approx$ 1.00). Shapley Additive Explanations (SHAP) reveal that the model's predictions align with physical expectations by distributing influence across meaningful features such as capacity and voltage, thus improving interpretability and trust.

The proposed framework demonstrates a powerful integration of proximity-based modeling and explainable machine learning, advancing the state-of-the-art in reliable SoC estimation across varying conditions. Additional subsampling and noise robustness analyses confirm that the framework remains stable under reduced data availability and sensor perturbations, ensuring practical reliability.

1. Introduction

Li-ion batteries (LIBs) have emerged as a preferred energy storage solution due to their favorable characteristics such as elevated energy density, low mass, extended operational life, and superior efficiency [1,2]. Over the last decade, active materials and structural configurations in Li-ion batteries have been significantly improved. These enhancements have increased the specific energy from nearly 150 Wh/kg to approximately 250 Wh/kg. This progress has been pivotal in facilitating the shift from conventional ICE vehicles to electric alternatives. Nonetheless, the driving range of electric vehicles (EVs) continues to fall short of that offered by internal combustion engine (ICE) vehicles, owing to intrinsic limitations in energy density, insufficient charging infrastructure, and progressive battery covering [3,4]. Current investigations target the development of solid-state electrolytes, high-capacity cathodic materials, and accelerated charging mechanisms to overcome these challenges and bridge the performance gap between EVs and traditional vehicles [5]. Concurrently, improvements in Battery Management Systems (BMS) have reinforced the operational stability, safety, and overall functionality of LIBs [6]. Contemporary BMS architectures incorporate sophisticated control algorithms that govern the behavior of individual cells within the battery pack, effectively averting conditions such as thermal incidents and voltage imbalances [7]. As BMS technologies evolve, accurate quantification of critical indicators like State of Health (SoH) and State of Charge (SoC) has become central to performance optimization and user reliability [8].

Historically, the SoC has been estimated using coulomb counting and open-circuit voltage (OCV) analysis, often with Kalman Filter-based

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Nomenclature

AI	Artificial Intelligence	x	Battery data sample (observation)
ANN	Artificial Neural Network	$\Phi(x)$	Descriptive feature vector of x
BMS	Battery Management System	$\phi_i(x)$	ith feature of x
CC–CV	Constant Current–Constant Voltage	$I(t)$	Discharge current at time t (mA)
CNN	Convolutional Neural Network	V	Voltage (V)
CV	Cross-Validation	C_{actual}	Actual battery capacity (mAh)
DBSCAN	Density-Based Spatial Clustering of Applications with Noise	E	Energy (mWh)
DPLM	Descriptive Proximity-based LightGBM Model	T	Ambient temperature (°C)
DoD	Depth of Discharge	$SoC(t)$	State of charge at time t (%)
EV	Electric Vehicle	η_i	Coulombic efficiency
LIB	Li-ion Battery	t_0, t	Initial and current time (s)
LightGBM	Light Gradient Boosting Machine	δ_Φ	Descriptive proximity relation
LSTM	Long Short-Term Memory	$Q(A)$	Set of descriptions for subset A
MAD	Median Absolute Deviation	$\mu_R(A,B)$	Fuzzy proximity between sets A and B
MAE	Mean Absolute Error	$\|\cdot\|_2$	Euclidean norm
ML	Machine Learning	ϵ	Distance threshold for DBSCAN
OCV	Open Circuit Voltage	n	Number of samples
PDP	Partial Dependence Plot	y_i	True SoC value of sample i (%)
R^2	Coefficient of Determination	$\hat{y}_i$	Predicted SoC value of sample i (%)
RF	Random Forest	$\bar{y}$	Mean of true SoC values (%)
RMSE	Root Mean Squared Error	c_i	Cluster label of x_i
RNN	Recurrent Neural Network		
SHAP	SHapley Additive exPlanations		
SoC	State of Charge		
SoH	State of Health		
SVR	Support Vector Regression		
XAI	Explainable Artificial Intelligence		
XGBoost	Extreme Gradient Boosting		

methods [9]. However, these traditional techniques lose accuracy as the battery ages and temperature conditions change, which undermines predictive reliability [10]. To mitigate such issues, machine learning (ML) algorithms such as support vector machines (SVM), artificial neural networks (ANN), and deep learning models-are increasingly being adopted. These approaches excel at capturing nonlinear patterns and adapting to dynamic operational contexts, thereby enhancing the reliability of SoC and SoH estimations [11].

A substantial body of research has focused on enhancing the estimation techniques for battery SoH and SoC, resulting in a range of innovative methodologies. Dineva et al. utilized machine learning algorithms, trained with suitable datasets, to estimate the SoC of Li-ion batteries across varying C-rate conditions. Beyond experimental validation, they developed a battery model capable of simulating fluctuating load profiles, demonstrating its effectiveness in predicting voltage and SoC under dynamic operating conditions [12]. Another study employed an aging-based experimental approach combined with a Kalman filter algorithm to assess capacity degradation and integrate SoH with SoC estimations. The proposed method was validated through physical testing, and the resulting estimation errors were quantified to confirm its practical viability [13]. In the related work, Liu et al. introduced a Kalman filter-based framework for precise SoC tracking, aimed at preventing battery overcharge and deep discharge scenarios. The model incorporated real-time parameter identification through a least squares approach and achieved an SoC estimation accuracy within a 2% error margin under diverse testing conditions [14]. Furthermore, Yang proposed a deep learning solution by designing a hybrid convolutional neural network to forecast the remaining useful life and total cycle lifespan of Li-ion batteries. The approach featured a cycle attention mechanism, leveraging input features such as voltage, current, and temperature to enhance predictive accuracy [15].

Various artificial intelligence (AI) strategies have been actively integrated by the research team to enable robust SoH prediction, incorporating deep learning architectures, explainable AI (XAI) methodologies, and hybrid neural network models. Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) have been implemented successfully, demonstrating notable predictive performance in domains such as unmanned aerial systems and electric mobility [16, 17]. In particular, CNN-RNN hybrid models have yielded significant accuracy when applied to time-series inputs such as voltage and temperature profiles, especially under dynamic load conditions. To further enhance estimation accuracy, ensemble-based learning frameworks, namely gradient boosting, XGBoost, LightGBM, and model stacking-have been utilized. These approaches outperform traditional statistical models in many settings. They capture nonlinear degradation dynamics and adapt effectively to a wide range of operational conditions [17]. Quantitative assessments have demonstrated improvements of approximately $30-45\%$ in root mean square error (RMSE) and mean absolute error (MAE) compared to classical methods such as polynomial regression and ARIMA.

Long Short-Term Memory (LSTM) and Bidirectional LSTM (Bi-LSTM) networks have been employed to address the time-dependent nature of degradation processes. These architectures have shown particular efficacy in modeling long-range temporal dependencies within features such as charge/discharge current patterns and internal resistance trends. Bi-LSTM models, in particular, have exhibited enhanced reliability in early-stage anomaly detection, especially in systems subjected to variable thermal conditions and inconsistent operational regimes [18]. To promote transparency and trust in model predictions, the study prioritizes explainability. This is achieved by integrating SHAP (Shapley Additive Explanations) and Partial Dependence Plots (PDP). These XAI tools have facilitated the identification of critical input features-such as mean discharge voltage, temperature variance, and depth of discharge (DoD) thereby enabling root-cause analysis and the derivation of actionable insights [16]. Models augmented with

SHAP-based interpretability have shown improved explanatory power. In addition, they offer operational advantages when used in safety-critical battery management applications. Additionally, investigations have been conducted into the effects of thermal profiles, operational stress rates, and degradation modes on SoH modeling, thereby ensuring robust performance across diverse application scenarios [19]. Real-time monitoring capabilities and adaptive control strategies are also being developed as part of an integrated approach to intelligent battery management. By leveraging a comprehensive suite of AI-driven methodologies, significant advancements have been made in enhancing the functionality and reliability of battery management systems. These contributions support extended battery lifespans, improved operational efficiency, and alignment with broader objectives pertaining to sustainable and resilient energy storage technologies [20]. In addition, we conducted subsampling and noise robustness analyses to demonstrate that the proposed framework maintains stability under reduced data availability and sensor perturbations, further supporting its practical reliability.

Beyond its practical implications, the proposed approach is theoretically grounded in the descriptive proximity framework, which draws from the foundations of Efremović proximity and its generalizations [21]. The use of fuzzy proximity measures and feature-based neighborhoods fits naturally within the framework of proximal relator spaces. These are generalizations of proximity spaces that incorporate descriptive and algebraic relations [22,23]. These spaces support the modeling of mathematical objects through descriptive intersections, spatial isometries, and approximate algebraic operations [24–27]. The relator-based perspective further enables a topological interpretation of data structure, where the concept of descriptive nearness between clusters can be formalized using fuzzy relational operators that define soft boundaries and semantic overlap [28–30]. The inclusion of fuzzy proximity into the model thus bridges quantitative machine learning and qualitative relational reasoning, fostering a novel integration of data topology and predictive analytics.

Building upon this foundation, the framework of *approximately algebraic structures* within proximal relator spaces offers a mathematically rigorous and topologically coherent setting for analyzing uncertainty, vagueness, and partial memberships among data-driven clusters. These algebraic generalizations, which extend traditional group, ring, and semigroup concepts into descriptive and proximity-enriched domains [24,25,31,32], enable the representation of complex structural behaviors observed in real-world battery systems. In our context, the descriptive feature space endowed with fuzzy proximity metrics facilitates the construction of an *approximately algebraic structure*, wherein each cluster reflects an algebraic object exhibiting approximate closure and associativity with respect to similarity-based neighborhood operations. This reinforces the mathematical interpretability of the DBSCAN-derived clusters and suggests a formal basis for potentially embedding soft relations, such as intra and inter-cluster proximities, into the learning model in future extensions. Thus, the integration of descriptive proximity with approximately algebraic structures serves as a foundational bridge between symbolic algebra and data-driven learning, yielding both theoretical soundness and computational tractability in our proposed DPLM framework.

2. Theoretical background

This section introduces the foundational concepts and mathematical structures underpinning our proposed framework, including relator spaces, proximity relations, and fuzzy proximity. These concepts provide a theoretical basis for the descriptive and structural features utilized in our model.

Let X be a nonempty set. A collection $\mathcal{R}$ of relations on X is termed a *relator* [33]. The pair $(X, \mathcal{R})$ forms a relator space, which naturally generalizes uniform spaces [34]. When this collection comprises proximity relations, it defines a proximal relator space $(X, \mathcal{R}_\delta)$ [23]. As described in [23], $\mathcal{R}_\delta$ includes various proximity relations such as Efremovič proximity δ_E [21], Lodato proximity $\delta_{\mathcal{L}}$ [35], Wallman proximity δ_W and descriptive proximity δ_Φ, which defines $\mathcal{R}_{\delta_\Phi}$ [27,30].

Let us define a relation δ on the power set of a nonempty set X. The notation $A\delta B$ means that set A is proximal to set B, and $A\underline{\delta}B$ means that set A is not proximal to set B. For all $A, B, C \in \mathcal{P}(X)$, we consider the following axioms:

(A_0) $\emptyset\ \underline{\delta}\ A$.
(A_1) $A\ \delta\ B \Longrightarrow B\ \delta\ A$.
(A_2) $(A \cup B)\ \delta\ C \Longleftrightarrow A\ \delta\ C$ or $B\ \delta\ C$.
(A_3) $A\ \delta\ B \Longrightarrow A \neq \emptyset,\ B \neq \emptyset$.
(A_4) $A \cap B \neq \emptyset \Longrightarrow A\ \delta\ B$.

If it satisfies (A_0), (A_1), (A_2), (A_3) and (A_4), δ is called a *basic proximity* on X.

In this work, proximal relator spaces are consists of concrete points. A non-abstract point has a location and some features that can be measured [36].

Measurable features of an object can be represented using probe functions. A probe function $\phi : X \to \mathbb{R}$ maps each sample point within a given space or digital image to a real number, effectively quantifying specific characteristics of that sample. Consider $\Phi(x) = (\phi_1(x), \dots, \phi_n(x))$, which denotes a comprehensive description of the object x. Each $\phi_i(x)$ within this tuple represents a distinct measurable feature of x, allowing for a detailed representation of every element $x \in X$. This framework facilitates the analysis and comparison of objects based on their quantified features.

Definition 2.1 (*[37]*)**.** Let X be a nonempty set composed of concrete points, and let Φ be an object description. For any subset $A \subseteq X$, the set description of A is defined as

$$\mathcal{Q}(A) = \{\Phi(a) \mid a \in A\}.$$

Definition 2.2 (*[29,37]*)**.** Let X be a nonempty set of concrete points, and let A and B be any subsets of X. The descriptive intersection of A and B is given by

$$A \underset{\Phi}{\cap} B = \{x \in A \cup B \mid \Phi(x) \in \mathcal{Q}(A) \ \ and \ \ \Phi(x) \in \mathcal{Q}(B)\}.$$

Definition 2.3 (*[30]*)**.** Let X be a nonempty set consisting of concrete points, and let A and B be any subsets of X. If $\mathcal{Q}(A) \cap \mathcal{Q}(B) \neq \emptyset$, then A is said to be descriptively near B, denoted as $A\delta_\Phi B$. Conversely, if $\mathcal{Q}(A) \cap \mathcal{Q}(B) = \emptyset$, then A is described as being descriptively far from B, denoted as $A\underline{\delta_\Phi}B$.

Throughout this article, we focus on the descriptive basic proximity δ_Φ, which is sufficient for modeling and analysis in our application context.

Definition 2.4 (*[23,30]*)**.** A *descriptive basic proximity* δ_Φ on the power set $P(X)$ satisfies the following conditions for all $A, B, C \in P(X)$:

0° $\emptyset\ \underline{\delta_\Phi}\ A$.
1° $A\ \delta_\Phi\ B \Rightarrow B\ \delta_\Phi\ A$.
2° $A\ \delta_\Phi\ B \Rightarrow A \neq \emptyset$ and $B \neq \emptyset$.
3° $\mathcal{Q}(A) \cap \mathcal{Q}(B) \neq \emptyset \Rightarrow A\ \delta_\Phi\ B$.
4° $A\ \delta_\Phi\ (B \cup C) \Leftrightarrow A\ \delta_\Phi\ B$ or $A\ \delta_\Phi\ C$.

Definition 2.5 (*[22]*)**.** Let $(X, \mathcal{R})$ be a proximal relator space,

$$\begin{aligned} \mu_{\mathcal{R}} : \quad P(X) \times P(X) &\longrightarrow [0,1] \\ (A,B) &\longmapsto \mu_{\mathcal{R}}(A,B) \end{aligned}$$

be a fuzzy relation and $A, B \subset X$; then

$$\mathcal{R}_\mu = \left\{ \left((A,B), \mu_{\mathcal{R}}(A,B) \right) \mid A, B \in P(X) \right\}$$

is called a fuzzy proximity relation, provided it satisfies the following axioms:

For all $A, B, C \in P(X)$,

$\left(N_{\mu_{\mathcal{R}}}\right)_1$ $\mu_{\mathcal{R}}(A,\emptyset)=0.$

$\left(N_{\mu_{\mathcal{R}}}\right)_2$ $\mu_{\mathcal{R}}(A,B)=\mu_{\mathcal{R}}(B,A).$

$\left(N_{\mu_{\mathcal{R}}}\right)_3$ $\mu_{\mathcal{R}}(A,B)\neq 0$ implies $A\delta B$ for $\delta\in\mathcal{R}$.

$\left(N_{\mu_{\mathcal{R}}}\right)_4$ $\mu_{\mathcal{R}}(A,B\cup C)\neq 0$ implies $\mu_{\mathcal{R}}(A,B)\neq 0$ or $\mu_{\mathcal{R}}(A,C)\neq 0$.

The set of all fuzzy proximity relations on $P(X)$ is denoted by $\mathcal{P}_{\mu_{\mathcal{R}}}(X)$. Therefore, $\mu_{\mathcal{R}}(A,B)$ is called fuzzy proximity measure.

Definition 2.6 (*Fuzzy Proximity Between Clusters*). Let X be a set of battery discharge instances, and let $A,B\subseteq X$ be non-empty clusters formed in a normalized descriptive feature space via $\Phi : X\to\mathbb{R}^n$. The fuzzy proximity $\mu_{\mathcal{R}}(A,B)$ is defined as:

$$\mu_{\mathcal{R}}(A,B) = \begin{cases}\exp\left(-\frac{1}{|A|\,|B|}\sum_{x\in A}\sum_{y\in B}\|\Phi(x)-\Phi(y)\|_2\right), & A\neq\emptyset \text{ and } B\neq\emptyset\\ 0, & \text{otherwise}\end{cases} \quad (1)$$

This definition transforms the average pairwise Euclidean distance into a fuzzy similarity value in the interval $[0,1]$.

Theorem 1. *The function $\mu_{\mathcal{R}}$ given in Definition 2.6, is a valid fuzzy proximity relation over $P(X)$.*

Proof. Let $A,B,C\subseteq X$. We verify each axiom:

$\left(N_{\mu_{\mathcal{R}}}\right)_1$ By definition,

$$\mu_{\mathcal{R}}(A,\emptyset)=0=\mu_{\mathcal{R}}(\emptyset,B)\quad\text{for any } A,B\subseteq X.$$

$\left(N_{\mu_{\mathcal{R}}}\right)_2$

$$\mu_{\mathcal{R}}(A,B)=\exp\left(-\frac{1}{|A|\,|B|}\sum_{x\in A}\sum_{y\in B}\|\Phi(x)-\Phi(y)\|_2\right)$$

Since $\|\Phi(x)-\Phi(y)\|_2 = \|\Phi(y)-\Phi(x)\|_2$, the expression is symmetric in A and B. Hence:

$$\mu_{\mathcal{R}}(A,B)=\mu_{\mathcal{R}}(B,A)$$

$\left(N_{\mu_{\mathcal{R}}}\right)_3$ Suppose $\mu_{\mathcal{R}}(A,B)\neq 0$. Then A and B are non-empty, and there exists bounded proximity between their members. This implies that some relation $A\mathcal{R}B$ exists in the descriptive proximity space.

$\left(N_{\mu_{\mathcal{R}}}\right)_4$ Suppose $\mu_{\mathcal{R}}(A,B\cup C)\neq 0$, i.e., A is proximal to $B\cup C$. Then:

$$\exists\, x\in A, z\in B\cup C \text{ such that } \|\Phi(x)-\Phi(z)\|_2 \text{ is small.}$$

This implies either $\mu_{\mathcal{R}}(A,B)\neq 0$ or $\mu_{\mathcal{R}}(A,C)\neq 0$, and thus $A\mathcal{R}B$ or $A\mathcal{R}C$. □

3. Data exploration and preprocessing

3.1. Experimental setup and data collection

In this study, high-power cylindrical 18 650-type lithium-ion cells with a nominal capacity of 2.8 Ah were tested under controlled laboratory conditions to ensure reliable data quality. The experimental setup involved discharge tests performed at two ambient temperatures (0 °C and 25 °C) and four current rates ($1C$, $2C$, $3C$, and $4C$), enabling simultaneous evaluation of rate capability and temperature sensitivity. Charging was carried out using the constant current–constant voltage (CC–CV) protocol at $1C$, with the voltage limited to 4.2 V and terminated when the current decreased to $0.05C$, thus guaranteeing complete yet safe charging. All experiments were conducted using the Neware BTS4000-5V12 A Battery Test System. This system provides highly reliable and precise current and voltage control within a voltage window of 2.5 V–4.2 V, ensuring accurate and reproducible electrochemical measurements.

Based on the collected charge–discharge profiles, the State of Charge (SoC) was computed according to

$$\mathrm{SoC}(t)=\mathrm{SoC}(t_0)-\frac{1}{C_{\text{actual}}}\int_{t_0}^{t}\eta_i I(t)\,dt, \quad (2)$$

and the resulting values were used as high-fidelity inputs to the proposed descriptive proximity-based learning framework.

3.2. Exploratory data analysis

To gain an initial understanding of the dataset, we conducted a comprehensive statistical analysis on seven key features: current (I), voltage (V), capacity (C), energy (E), SoC, cycle number, and temperature (T). The descriptive statistics for each feature are summarized in Table 1.

The current was kept constant near −2500 mA across all samples, ensuring consistency during the discharge process. Notably, the voltage ranged from 2.50 to 4.09 V, reflecting typical lithium-ion discharge behavior. The SoC values covered the full scale from 0% to 100%, which is critical for developing a robust and generalizable prediction model.

To evaluate the linear dependencies among the variables, we computed the Pearson correlation matrix shown in Table 2.

A strong positive correlation is observed between voltage and SoC ($r = 0.948$), indicating that voltage is a reliable indicator of battery charge level. Similarly, capacity and energy are almost perfectly correlated with each other ($r = 0.999$), while both are inversely correlated with SoC. Notably, current is highly correlated with temperature ($r = 0.910$), since different temperature conditions were tested using distinct current levels.

These insights guide the feature selection for the descriptive proximity vector $\Phi(x)$, which will form the foundation for clustering and fuzzy similarity analysis in subsequent sections.

3.3. Parameter sensitivity analysis for DBSCAN clustering

To assess the stability and robustness of the clustering process, we conducted a parameter sensitivity analysis on the DBSCAN algorithm by varying both the neighborhood radius ε and the `min_samples` parameter. The goal was to determine whether our adaptively computed ε value aligns with regions of consistent cluster formation, and to evaluate the impact of manual parameter choices on clustering outcomes.

We randomly sampled 5000 feature vectors from the combined battery discharge dataset (0°C and 25°C), and applied DBSCAN across a grid of $\varepsilon \in \{0.9, 1.0, 1.1, 1.2, 1.3, 1.4\}$ and `min_samples` $\in \{10, 15, 20, 25, 30\}$. For each configuration, we recorded the number of clusters and the number of data points labeled as noise. The results are summarized in Table 3 and visualized in Fig. 1.

The heatmap in Fig. 1 reveals that the number of clusters remains highly stable (typically 3 clusters) for ε values in the range $[0.9, 1.1]$ across all tested `min_samples` values. However, for $\varepsilon \geq 1.2$, the algorithm consistently identifies only two clusters, regardless of the `min_samples` setting. This transition indicates a sensitivity boundary beyond which cluster granularity deteriorates due to over-smoothing in the feature space.

The results validate our choice of $\varepsilon_{\text{adaptive}} = 1.21$, computed via the MAD-based proximity calibration. This value lies at the transition region between over-clustering and under-clustering, enabling a balanced and data-driven detection of meaningful proximity structures. The near-perfect agreement between manually-tuned stable regions (around $\varepsilon = 1.1$) and the adaptive result strengthens the theoretical and empirical

Table 1
Descriptive statistics of the main variables.

Feature	Count	Mean	Std	Min	25%	50%	75%	Max
Current (mA)	67 428	−2500.80	0.32	−2501.86	−2501.11	−2500.74	−2500.37	−2499.25
Voltage (V)	67 428	3.51	0.29	2.50	3.34	3.53	3.75	4.09
Capacity (mAh)	67 428	1175.24	683.98	0.00	584.98	1170.66	1756.13	2512.28
Energy (mWh)	67 428	4320.42	2431.26	0.00	2247.57	4376.77	6381.78	8944.12
SoC (%)	67 428	49.99	28.88	0.00	24.99	50.00	75.01	100.00
Cycle	67 428	5.51	2.87	1.00	3.00	6.00	8.00	10.00
Temperature (°C)	67 428	13.28	12.48	0.00	0.00	25.00	25.00	25.00

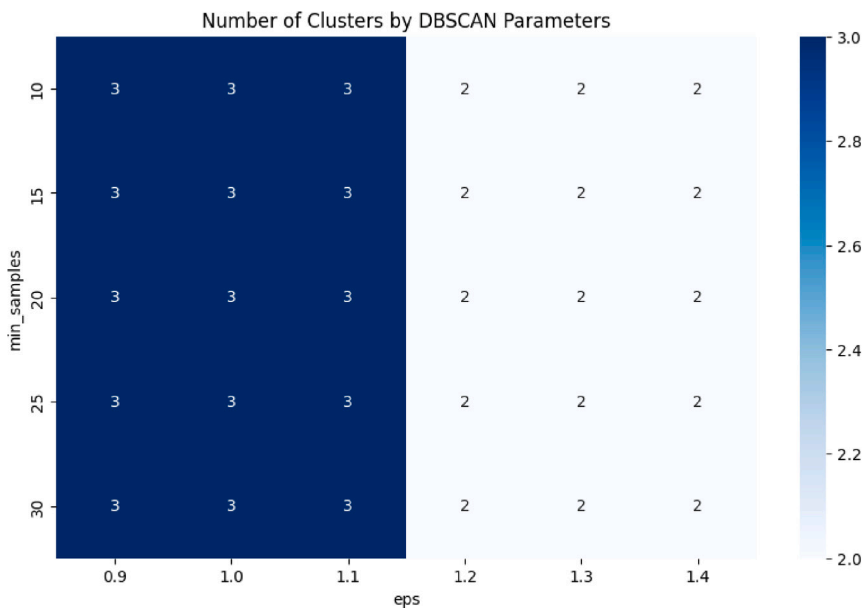

Fig. 1. Heatmap showing the number of clusters formed under different combinations of ε and `min_samples`.

Table 2
Pearson correlation coefficients among the features.

	Current	Voltage	Capacity	Energy	SoC	Cycle	Temp.
Current	1.000	0.127	0.119	0.147	−0.022	−0.031	0.910
Voltage		1.000	−0.926	−0.911	0.948	0.015	0.164
Capacity			1.000	0.999	−0.992	0.007	0.107
Energy				1.000	−0.987	0.009	0.137
SoC					1.000	−0.000	0.000
Cycle						1.000	0.001
Temperature							1.000

Table 3
Sensitivity of DBSCAN clustering to ε and `min_samples` values.

ε	`min_samples`	Number of clusters	Number of noise points
0.9	10	3	6
0.9	20	3	7
1.0	15	3	6
1.1	25	3	6
1.2	10	2	4
1.3	20	2	4
1.4	30	2	4

justification for our approach. Furthermore, the small number of noise points across all configurations suggests that the DBSCAN procedure is robust against minor variations in parameterization, supporting the reliability of our descriptive clustering pipeline.

To assess how this adaptive clustering decision influences downstream prediction tasks, we conducted an additional comparative experiment. To quantify the impact of ε selection on downstream SoC prediction, we retrained the LightGBM regression model using cluster labels derived from the fixed $\varepsilon = 1.1$ setting (with `min_samples` = 20), instead of the previously used adaptive threshold. Using the same random 5000-sample subset for fairness, the performance metrics were:

- RMSE: 0.189
- MAE : 0.146
- R^2: 1.000

These results are nearly identical to those achieved using the adaptive ε value ($\varepsilon_{\text{adaptive}} = 1.21$), which produced RMSE = 0.188 and MAE = 0.145. The negligible difference in error metrics further confirms the empirical stability of the clustering mechanism in our dataset, which consists of highly structured discharge curves under two distinct temperature conditions.

Although the battery discharge data comprises samples obtained under two distinct thermal conditions (0°C and 25°C), we deliberately chose not to stratify the analysis by temperature. This decision is theoretically grounded and empirically validated by the inclusion of temperature as a feature in the descriptive vectors $\Phi(x)$. Its influence is inherently embedded within the feature space and thus reflected in the clustering and prediction processes. Moreover, SHAP-based interpretability analysis revealed that temperature, while physically significant, had a limited direct impact on the model's output due to its correlation with more dominant features such as capacity and voltage. As such, the integrated modeling approach successfully captures temperature effects without requiring separate stratified modeling, thereby maintaining model generalizability and avoiding unnecessary fragmentation of the dataset.

In closing, although the adaptively determined threshold is theoretically more robust and better suited for generalization to other datasets, its practical performance on this specific dataset was closely matched by a well-tuned fixed value. Nonetheless, we retain the adaptive method in our proposed DPLM framework for its mathematical consistency, automation potential, and generalizability across more heterogeneous battery data.

Additionally, to test stability under realistic constraints, we performed robustness analyses involving subsampling and Gaussian noise injection. Details and results are reported in Section 5.5.

3.4. Adaptive ε-proximity via robust statistics

One of the critical parameters in proximity-based clustering algorithms, such as DBSCAN, is the selection of the distance threshold ε. A poorly chosen fixed ε may lead to over-clustering or under-segmentation, particularly in datasets characterized by heterogeneous density, such as battery discharge profiles under varying thermal and electrical conditions.

To address this issue, we adopted a robust statistical method for determining an adaptive ε, grounded in the *Median Absolute Deviation (MAD)*. Let D denote the set of pairwise Euclidean distances between a random sample of $n = 3000$ normalized feature vectors $\Phi(x) \in \mathbb{R}^6$, defined as:

$$D = \left\{ \left\| \Phi(x_i) - \Phi(x_j) \right\|_2 : x_i, x_j \in X_{\text{sample}},\ i < j \right\} \tag{3}$$

Then, the MAD is computed as:

$$\text{MAD} = \text{median}\left(|d - \text{median}(D)| : d \in D\right) \tag{4}$$

The adaptive proximity threshold is defined by the scaled MAD:

$$\varepsilon_{\text{adaptive}} = k \cdot \text{MAD}, \quad \text{where } k = 1.4826 \tag{5}$$

This scaling factor ensures consistency with the standard deviation under the assumption of normally distributed distances. The computed value for the adaptive proximity threshold was:

$$\varepsilon_{\text{adaptive}} = 1.21 \tag{6}$$

This adaptively determined ε will be employed in the subsequent DBSCAN clustering to identify descriptive proximity-based groups within the battery discharge data. The approach ensures that clusters are sensitive to local variations in feature space density, thereby improving the robustness and interpretability of the resulting model.

4. Descriptive clustering and fuzzy modeling

4.1. Descriptive clustering using adaptive ε-proximity

Given the complexity and heterogeneity of battery discharge behavior across varying temperatures and current rates, traditional clustering methods that rely on a fixed distance threshold ε are often insufficient. To address this limitation, we applied an adaptive ε-proximity framework rooted in robust statistics, as described in the previous section.

In this setting, each battery measurement $x \in X$ is represented by a normalized feature vector $\Phi(x) \in \mathbb{R}^6$, encompassing current, voltage, capacity, energy, cycle count, and temperature. These vectors form the basis of the descriptive proximity space.

Computational Constraints and Sampling Strategy. The original dataset comprises over 67,000 observations, rendering full pairwise distance computation infeasible due to excessive memory requirements. To circumvent this challenge without compromising statistical integrity, we employed a sampling-based approach. Specifically, a random subset of 5000 normalized vectors was selected, preserving the overall distribution of the data.

DBSCAN Clustering with Adaptive ε. DBSCAN algorithm was applied to the sampled dataset using the adaptively determined threshold $\varepsilon_{\text{adaptive}} = 1.21$ and `min_samples` = 20. This proximity-based clustering technique is well-suited for descriptive spaces, as it identifies dense regions in feature space without requiring prior knowledge of the number of clusters.

Table 4
Cluster assignment using DBSCAN on a random sample ($n = 5000$).

Cluster label	Number of points
Cluster 0	2689
Cluster 1	2309
Noise (−1)	2

The resulting cluster distribution is presented in Table 4.

The clusters C_0 and C_1, identified through DBSCAN using adaptive ε-proximity, represent distinct operational regimes in the battery discharge dataset. Rather than being arbitrary numerical labels, these clusters encapsulate meaningful patterns emerging from variations in thermal and electrical conditions. The clustering algorithm detected two high-density regions in the descriptive feature space-formed by combinations of current, voltage, capacity, energy, temperature, and cycle count-that correspond to empirically distinct behavioral modes.

Cluster C_0 primarily comprises samples obtained under more demanding conditions, such as low ambient temperature (e.g., 0 °C) and higher discharge currents. These instances typically exhibit steeper voltage drops and more aggressive SoC transitions due to increased internal resistance and electrochemical sluggishness. In contrast, cluster C_1 predominantly captures nominal or favorable operating states (e.g., 25 °C, moderate current), where the battery maintains a stable discharge profile with smoother energy consumption.

From a practical standpoint, C_0 can be associated with suboptimal, stress-induced regimes, while C_1 reflects efficient, baseline performance. The inclusion of cluster labels as a categorical feature in the SoC prediction model allows the learning algorithm to distinguish between these regimes, even when other numerical features overlap. Although the SHAP analysis indicated that the explicit contribution of the `Cluster` variable was modest, its presence facilitated a better-structured input space that enhanced the model's overall performance and generalizability.

This semantic interpretation of C_0 and C_1 provides operational insight beyond mere statistical grouping, thereby supporting the hybrid framework's emphasis on interpretable and context-aware battery analytics.

The outcome demonstrates a high degree of structure in the dataset, with over 99.9% of the sampled instances assigned to meaningful clusters. Only two points were considered outliers, indicating that the chosen adaptive ε value effectively captures the underlying distribution of the discharge patterns.

This result validates the descriptive proximity assumptions and confirms the feasibility of building approximately vector subspaces within each cluster. These clusters now serve as the foundation for subsequent fuzzy proximity analysis and explainable SoC estimation.

4.2. Fuzzy proximity among descriptive clusters

Following the identification of descriptive clusters using adaptive ε-proximity, we now assess the degree of fuzzy similarity among them. This analysis is essential to model gradual transitions and uncertainties in the feature space, particularly relevant for SoC estimation under varying operational conditions.

Definition 4.1 (*Fuzzy Proximity Between Clusters*). Let $A, B \subseteq X$ be two clusters in the normalized feature space, where each element $x \in A \cup B$ is represented by a descriptive feature vector $\Phi(x) \in \mathbb{R}^n$. The fuzzy proximity $\mu_{\mathcal{R}}(A, B)$ between these clusters is defined by the exponential decay of their average pairwise Euclidean distance:

$$\mu_{\mathcal{R}}(A, B) = \exp\left(-\frac{1}{|A|\,|B|} \sum_{x \in A} \sum_{y \in B} \|\Phi(x) - \Phi(y)\|_2 \right) \tag{7}$$

Table 5
Fuzzy proximity values $\mu_{\mathcal{R}}(A, B)$ among descriptive clusters. Diagonal entries represent **intra-cluster proximity**, while off-diagonal entries represent **inter-cluster proximity**.

	C_0	C_1
C_0 (intra)	**0.0657**	0.0207
C_1 (inter)	0.0207	**0.0908**

This formulation satisfies the key intuition of fuzzy proximity: the closer two clusters are in the feature space, the higher their similarity score. It differs from classical proximity metrics by incorporating a continuous similarity spectrum that smoothly decays with increasing descriptive distance.

Definition 4.2 (*Intra-cluster Fuzzy Proximity*)**.** Let $C \subset X$ be a descriptive cluster, where each element $x \in C$ is represented by a descriptive feature vector $\Phi(x) \in \mathbb{R}^n$. The intra-cluster fuzzy proximity $\mu_{\mathcal{R}}^{\text{intra}}(C)$ is defined as:

$$\mu_{\mathcal{R}}^{\text{intra}}(C) = \exp\left(-\frac{1}{|C|^2}\sum_{x\in C}\sum_{y\in C}\|\Phi(x)-\Phi(y)\|_2\right)$$

This quantity measures the internal cohesion of cluster C in the descriptive feature space.

Definition 4.3 (*Inter-cluster Fuzzy Proximity*)**.** Let $C_i, C_j \subset X$ be two distinct descriptive clusters, where $C_i \cap C_j = \emptyset$. The inter-cluster fuzzy proximity $\mu_{\mathcal{R}}^{\text{inter}}(C_i, C_j)$ is defined as:

$$\mu_{\mathcal{R}}^{\text{inter}}(C_i, C_j) = \exp\left(-\frac{1}{|C_i \parallel C_j|}\sum_{x\in C_i}\sum_{y\in C_j}\|\Phi(x)-\Phi(y)\|_2\right)$$

This value quantifies the average descriptive similarity between samples in clusters C_i and C_j.

Importantly, even the self-proximity $\mu_{\mathcal{R}}^{\text{intra}}(C, C)$ is not necessarily equal to 1, since it depends on the internal dispersion of elements within cluster C. The closer and more homogeneous the elements are, the higher the value of $\mu_{\mathcal{R}}^{\text{intra}}(C, C)$; conversely, higher internal variance results in lower self-similarity. This makes the fuzzy proximity measure $\mu_{\mathcal{R}}$ particularly suitable for analyzing structural cohesion in descriptive clustering.

Note that each entry $\mu_{\mathcal{R}}(A, B)$ in Table 5 quantifies the average proximity between *all individual samples* in cluster A and cluster B, based on their descriptive feature vectors. In particular: - $\mu_{\mathcal{R}}^{\text{intra}}(C_0)$ denotes the average self-similarity of samples within cluster C_0, - $\mu_{\mathcal{R}}^{\text{inter}}(C_0, C_1)$ denotes the average proximity between samples in cluster C_0 and cluster C_1.

In our case, Cluster C_1 exhibited stronger internal proximity ($\mu_{\mathcal{R}}^{\text{intra}}(C_1) = 0.0908$) compared to Cluster C_0 ($\mu_{\mathcal{R}}^{\text{intra}}(C_0) = 0.0657$), reflecting its tighter concentration in the proximity-enhanced feature space. The inter-cluster proximity between C_0 and C_1 was notably lower ($\mu_{\mathcal{R}}^{\text{inter}}(C_0, C_1) = 0.0207$), indicating well-separated clusters (see Fig. 2) .

The highest fuzzy proximity is observed within Cluster C_1, indicating strong internal cohesion. In contrast, the inter-cluster similarity between C_0 and C_1 is notably low, reflecting clear separability in the descriptive feature space. These findings validate the robustness of the proximity-driven clustering process and provide a fuzzy relational foundation for subsequent SoC prediction and explanation.

5. SoC estimation framework

5.1. Algorithmic summary

To enhance reproducibility and provide a transparent overview of the computational pipeline, we present Algorithm 1, which outlines the core stages of our DPLM (Descriptive Proximity-based LightGBM Model). The algorithm includes data preprocessing, adaptive

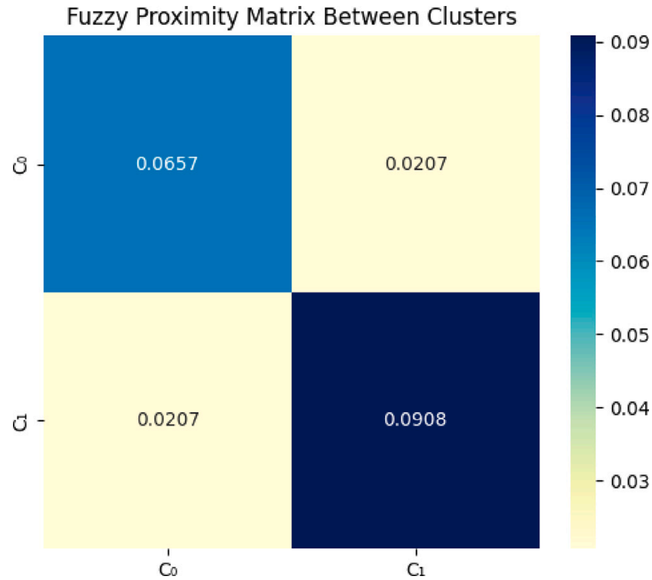

Fig. 2. Heatmap of fuzzy proximity values between descriptive clusters. Strong intra-cluster cohesion (dark diagonal) and low inter-cluster similarity (low intensity off-diagonal) suggest that Cluster C_0 and Cluster C_1 represent well-separated operating conditions.

ε-proximity estimation via robust statistics, proximity-based clustering using DBSCAN, fuzzy similarity computation between clusters, and the training of a LightGBM regressor for SoC prediction. By structuring the hybrid framework in a formal algorithmic format, we offer a concise, modular, and reproducible reference for researchers interested in proximity-aware battery modeling and interpretable machine learning.

Following the algorithmic pipeline, we proceeded to evaluate the predictive performance of the trained model using standard regression metrics such as RMSE, MAE, and R^2. To gain deeper insight into the model's internal reasoning, we applied SHAP-based interpretability techniques. These techniques allowed us to quantify the contribution of each input feature to the SoC prediction, thus bridging the gap between predictive accuracy and model transparency. The results of this interpretability analysis are presented in the following sections.

The diagram in Fig. 3 illustrates the full computational pipeline of our proposed DPLM framework. The process begins with battery discharge data, followed by normalization and removal of incomplete samples. An adaptive proximity threshold ε is computed using the Median Absolute Deviation (MAD) of pairwise distances among randomly sampled feature vectors. This threshold guides DBSCAN clustering to uncover descriptive structure in the data. Cluster labels are then used to construct approximately vector subspaces, and fuzzy proximity values are computed to capture inter-cluster similarity.

These descriptive labels are incorporated as features into the LightGBM regression model, which is trained to predict SoC. Finally, the model is evaluated using standard regression metrics, and SHAP values are computed to interpret the importance and influence of each feature. This workflow ensures both high accuracy and explainability in SoC estimation across varied operational conditions.

5.2. SoC estimation via proximity-enhanced gradient boosting

To estimate the SoC of lithium-ion batteries under varying thermal and electrical conditions, we employed a regression model based on the Light Gradient Boosting Machine (LightGBM) algorithm. This tree-based ensemble method is well-suited for tabular data, combining high predictive performance with low computational complexity.

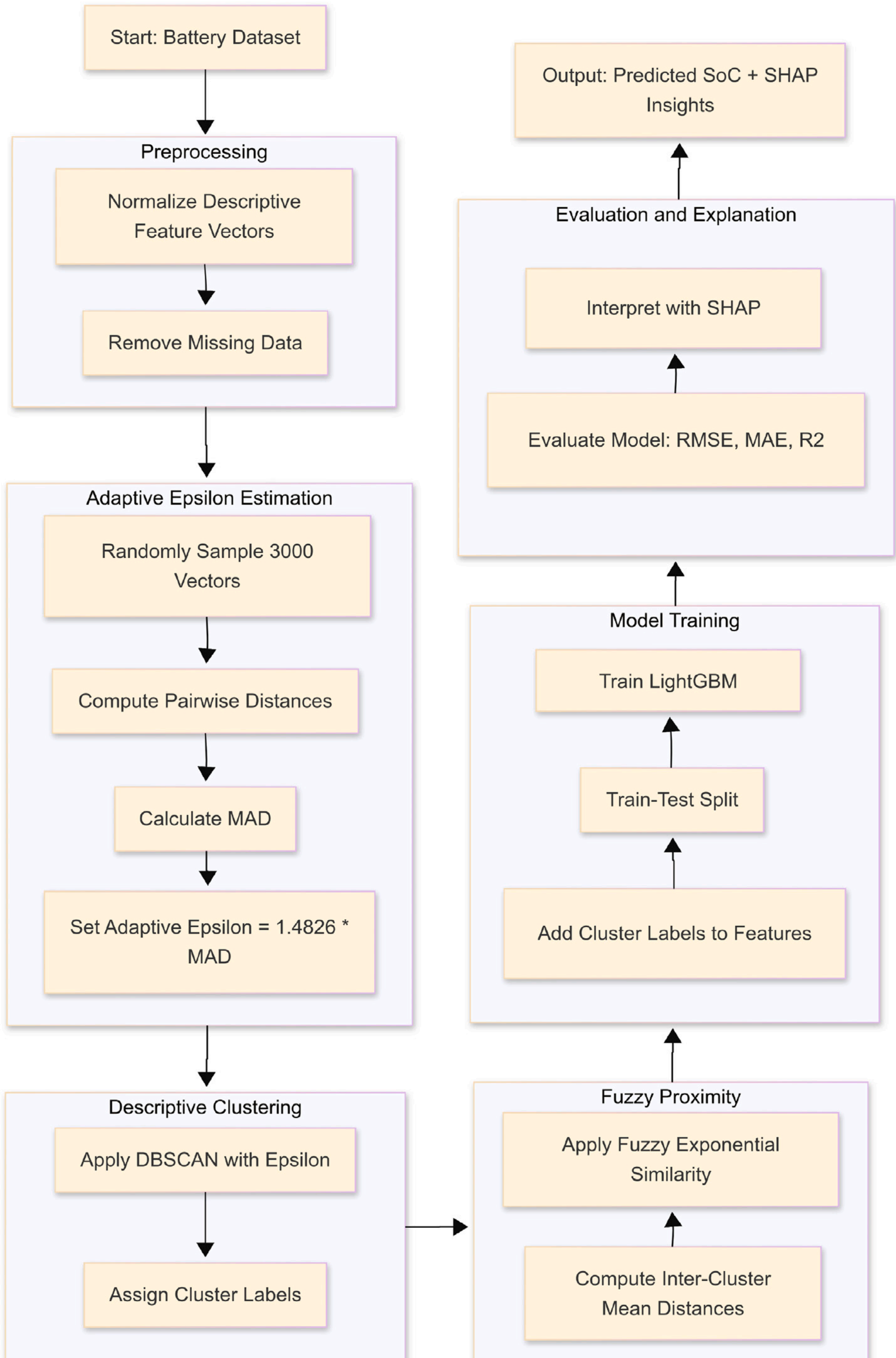

Fig. 3. Workflow diagram of the proposed DPLM (Descriptive Proximity-based LightGBM Model) for SoC estimation.

Each battery discharge instance was represented by a descriptive feature vector $\Phi(x)$ containing the following input variables:

- Current (mA)
- Voltage (V)
- Capacity (mAh)
- Energy (mWh)
- Cycle number
- Ambient Temperature (°C)
- Cluster label (obtained via adaptive DBSCAN over $\Phi(x)$)

The inclusion of the descriptive cluster label is particularly important, as it encodes latent operating regimes (e.g., low-temperature high-load or nominal discharge) uncovered through proximity-based density clustering. These regimes help structure the feature space and enhance predictive generalization.

To train and evaluate the model, the dataset was first preprocessed by removing noise-labeled samples identified through DBSCAN. A standard 80%–20% train–test split was applied. The LightGBM model was trained with default hyperparameters, and its predictive performance was evaluated using three standard regression metrics:

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{8}$$

Algorithm 1: DPLM: Descriptive Proximity-based LightGBM Model for SoC Estimation

Input: Battery discharge dataset $X = \{x_i\}_{i=1}^n$, with features $\Phi(x_i) = [I_i, V_i, C_i, E_i, N_i, T_i]$
Output: Trained LightGBM model $\mathcal{M}$ for predicting SoC

Step 1: Preprocessing
Normalize feature vectors $\Phi(x_i)$ for all $x_i \in X$.
Remove rows with missing values.
Step 2: Adaptive Proximity Threshold
Randomly sample m feature vectors from X.
Compute all pairwise Euclidean distances $D = \{\|\Phi(x_i) - \Phi(x_j)\|\}$.
Calculate the median absolute deviation:

$$\mathrm{MAD} = \mathrm{median}(|d - \mathrm{median}(D)|), \quad d \in D$$

Set the adaptive proximity threshold:

$$\varepsilon = 1.4826 \cdot \mathrm{MAD}$$

Step 3: Descriptive Clustering
Apply DBSCAN to $\Phi(X)$ using ε as the neighborhood radius.
Assign cluster label c_i to each sample $x_i \in X$.
Step 4: Fuzzy Proximity Matrix
For each pair of clusters (A, B), compute:

$$\mu_{\mathcal{R}}(A, B) = \exp\left(-\frac{1}{|A||B|}\sum_{x\in A}\sum_{y\in B}\|\Phi(x) - \Phi(y)\|_2\right)$$

Step 5: SoC Model Training
Append cluster labels c_i as an additional feature.
Split the dataset into training and test sets.
Train LightGBM model $\mathcal{M}$ on the training set to predict SoC.

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \tag{9}$$

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} \tag{10}$$

Evaluation on the test set demonstrated that the model achieved excellent predictive performance. The RMSE was 0.188, the MAE was 0.145, and the coefficient of determination (R^2) reached 1.000. These values indicate near-perfect SoC estimation accuracy and confirm the model's ability to capture complex nonlinear relationships in the descriptive feature space.

The integration of structural information through proximity-based clustering proved particularly valuable, enabling the model to distinguish between subtle operational variations that might otherwise remain hidden in raw sensor features. This modeling foundation sets the stage for a more detailed interpretability analysis, which is presented in the following section using SHAP.

5.3. Generalization assessment via cross-validation

To evaluate the robustness and generalizability of the proposed DPLM framework, we employed a 5-fold cross-validation (CV) strategy. Unlike a simple 80%-20% train–test split, cross-validation provides a more statistically reliable assessment of model performance by ensuring that every data point is used for both training and validation.

Battery discharge data often exhibit variability due to temperature, current load, and cycle count. A model that performs well on a specific train–test split may not generalize equally well across unseen operational scenarios. Thus, we applied 5-fold CV to test the model's consistency under varied data partitions. This technique helps quantify the stability of the learning process and mitigates the risks of overfitting or data-specific bias.

Table 6
5-Fold cross-validation performance of the DPLM model.

Metric	Mean	Standard deviation	Interpretation
RMSE	0.134	0.001	Extremely low average prediction error
MAE	0.108	0.001	High predictive consistency
R^2	1.000	0.000	Near-perfect variance explanation

To implement this validation strategy, the dataset was partitioned into five equal folds. In each iteration, four folds were used for training, and the remaining one for validation. The process was repeated five times so that each fold served as a test set exactly once. We used the following evaluation metrics:

- **RMSE:** Sensitive to large deviations; indicates overall prediction accuracy.
- **MAE:** Captures average absolute deviation; reflects typical error magnitude.
- R^2: Measures explained variance; ideal value is 1.

The outcomes of this procedure are summarized in the results table. Table 6 presents the average and standard deviation of each metric across the five folds.

The very low RMSE and MAE values indicate high precision in SoC predictions across all folds. The absence of variability in the R^2 scores underscores that the model explains nearly all the variance regardless of the test set used. These results confirm that the DPLM model not only learns the complex nonlinear mappings between battery features and SoC, but also maintains its accuracy across diverse operational conditions.

By demonstrating excellent performance in a fold-wise manner, the model proves to be robust and ready for deployment in real-world battery management systems, where generalization across multiple usage scenarios is critical.

5.4. Visual validation of SoC predictions

To complement the quantitative metrics of accuracy, we provide graphical validation of the model's predictive performance.

As demonstrated in Fig. 5, the proposed model exhibits near-ideal prediction performance, as evidenced by the symmetric, unimodal distribution of prediction errors centered around zero. The narrow bandwidth of the error distribution provides empirical evidence of the model's consistency and generalization capacity across the test set. This observation is further reinforced by the parity plot in Fig. 4, where the predicted SoC values are tightly clustered along the diagonal line representing perfect prediction.

These results are of particular significance for EV applications, where accurate and robust SoC estimation is critical for range prediction, battery health monitoring, and energy management. A near-perfect alignment between predicted and actual SoC ensures optimal utilization of battery capacity without overestimation, which could otherwise lead to range anxiety, or underestimation, which may cause premature charging and reduce battery lifespan. Furthermore, the low prediction error variance highlights the model's reliability under varying operational conditions-such as temperature fluctuations and discharge rates—making it well-suited for real-time onboard battery management systems in EVs. By leveraging the descriptive structure and fuzzy relational patterns within the data, the proposed model provides a mathematically grounded and interpretable SoC prediction framework with direct applicability to modern electric mobility ecosystems.

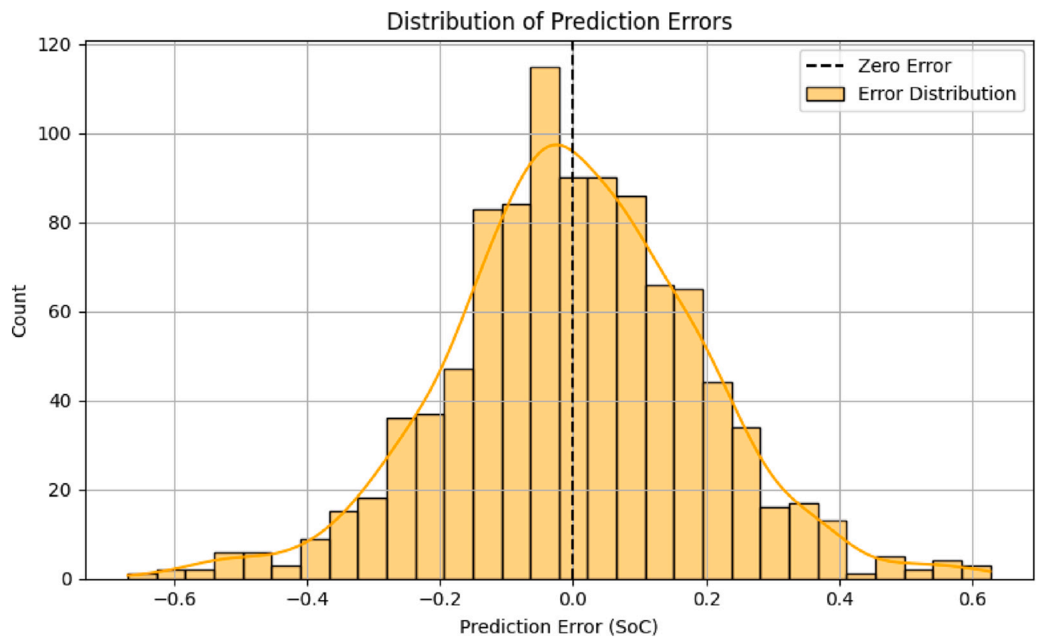

Fig. 4. Parity plot illustrating the alignment between actual and predicted SoC values. The near-perfect alignment along the identity line confirms the high precision of the proposed DPLM model.

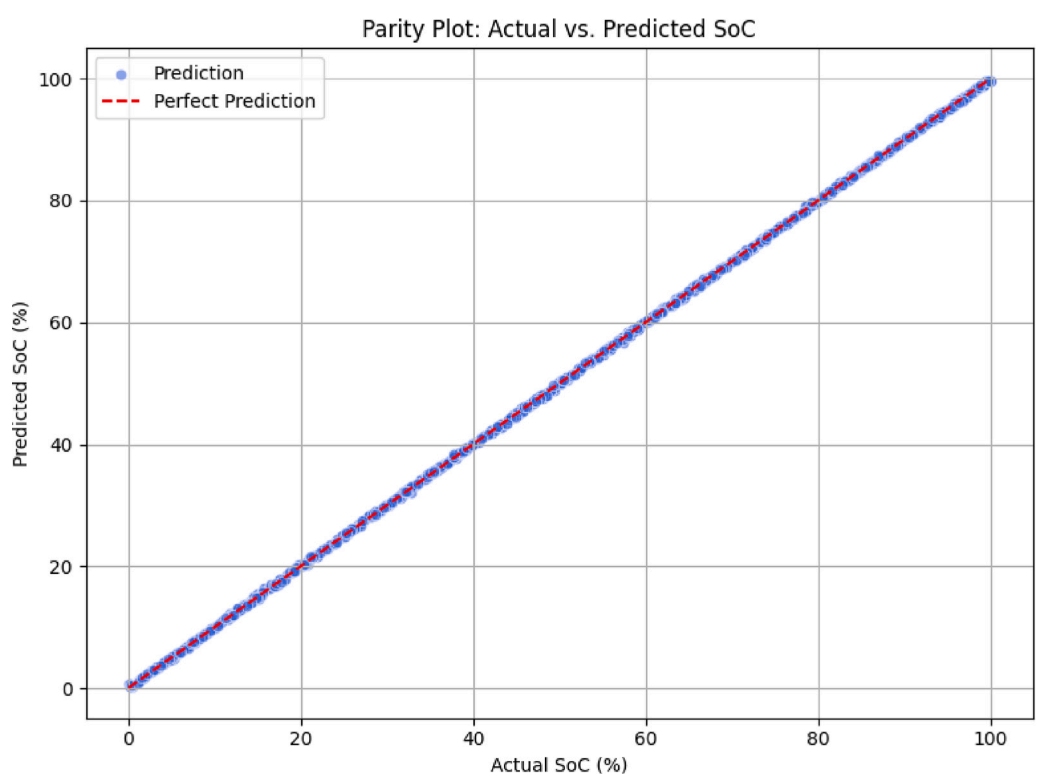

Fig. 5. Histogram of prediction errors for SoC estimation. The distribution is approximately centered around zero, indicating the absence of systematic bias. The narrow spread further reflects the model's robustness and low variance.

5.5. Robustness verification of DPLM

To further strengthen the experimental verification of the proposed DPLM framework, two robustness analyses were conducted: subsampling robustness and noise robustness. These tests address potential concerns regarding model stability under reduced data availability and resilience to sensor noise, respectively. Together, they provide compelling evidence that the proposed method is not only accurate but also generalizable and reliable in realistic operating scenarios.

5.5.1. Subsampling robustness

In practical applications, the availability of large-scale, high-quality datasets is not always guaranteed. A robust SoC estimation method

Table 7
Subsampling robustness results of DPLM. Performance remains stable across different training fractions, confirming generalization ability.

Sample fraction	RMSE	MAE	R^2
0.5	0.1886	0.1455	0.999959
0.7	0.2084	0.1596	0.999947
1.0	0.1927	0.1501	0.999954

Table 8
Noise robustness results of DPLM. Even under strong noise perturbations, the model maintains high accuracy and near-perfect R^2.

Noise Std Dev	RMSE	MAE	R^2
0.00	0.1830	0.1418	0.999961
0.01	0.3435	0.2575	0.999863
0.05	0.3806	0.2292	0.999832
0.10	0.4240	0.2298	0.999792

should maintain consistent accuracy even when trained on smaller subsets of the available data. To verify this, we conducted a subsampling analysis.

Method: The dataset was randomly downsampled to 50%, 70%, and 100% of its original size, with distinct random seeds to ensure diversity across subsets. The complete DPLM pipeline, including adaptive DBSCAN clustering, fuzzy proximity computation, and LightGBM regression, was applied to each subsample. Model performance was then evaluated using RMSE, MAE, and R^2 metrics.

As shown in Table 7, the predictive accuracy of DPLM remained consistently high across all subsampling levels. Even when trained on only 50% of the data, the model achieved nearly identical performance to the full dataset. This result demonstrates that the descriptive proximity-enhanced feature space provides strong structural information, enabling DPLM to generalize effectively under reduced data availability.

5.5.2. Noise robustness

In real-world scenarios, battery measurements are often subject to sensor noise and external disturbances. A reliable SoC estimator must therefore remain stable even when feature values are corrupted by random fluctuations. To assess this, we performed a noise robustness analysis.

Method: Gaussian noise with increasing standard deviations ($\sigma = 0.01, 0.05, 0.10$) was injected into the input features prior to normalization. The complete DPLM pipeline was retrained under each noisy condition, and performance was measured using the same regression metrics.

Table 8 shows that while error metrics increase under stronger noise perturbations, the R^2 score consistently remains above 0.9997. This confirms that DPLM is not only highly accurate under clean conditions but also resilient to noisy sensor inputs, ensuring robustness in real-world battery management systems.

The combined results of subsampling and noise robustness tests provide strong evidence that the proposed DPLM framework maintains accuracy, stability, and interpretability under challenging conditions. By demonstrating resilience to both reduced data availability and sensor noise, DPLM establishes itself as a reliable and generalizable solution for State-of-Charge estimation in practical battery applications.

6. Interpretability and explainability

To deepen the interpretability assessment of the proposed **Descriptive Proximity Learning Model (DPLM)**, we performed a comparative SHAP analysis against a conventional LightGBM model trained on the same dataset without the proximity-based enhancements. This section presents both visual and quantitative comparisons to elucidate the impact of descriptive clustering on model behavior.

6.1. Interpretability analysis via SHAP

To investigate the internal decision-making process of the trained SoC prediction model, we employed SHAP. SHAP provides local and global interpretability by attributing each prediction to individual feature contributions, grounded in cooperative game theory.

We used the TreeExplainer variant of SHAP, optimized for tree-based models like LightGBM. Fig. 6 shows both the mean absolute SHAP values and the detailed distribution of SHAP values across features.

The bar chart indicates that `CapaCitymAh` is the most influential feature, followed by `VoltageV` and `EnergymWh`. These features reflect the energy discharged during battery usage and are thus directly tied to the residual SoC.

Beyond feature ranking, visualizing individual impact provides deeper insights. The color-coded scatter plot reveals how each feature's value affects the model output. For instance, high values of `CapaCitymAh` (in red) typically lead to increased SoC predictions, while lower values (in blue) have the opposite effect. This aligns with the physical intuition that higher discharge capacity correlates with greater remaining energy.

The role of structural features is also noteworthy. Interestingly, the `Cluster` feature contributed modestly to the model's predictions, yet it played a crucial role in structuring the feature space. Its inclusion indirectly enhanced predictive power by enabling the model to differentiate among operational regimes.

Taken together, these results highlight that our proximity-based modeling approach is not only accurate but also interpretable. Such transparency is essential for applications involving safety-critical systems like electric vehicles or aerospace batteries.

7. Discussion of results and practical implications

This study proposed a novel hybrid framework for battery SoC estimation that integrates descriptive proximity-based clustering with gradient-boosted regression models, enhanced through explainable AI techniques. Experiments using real-world battery discharge data across different temperature and current settings demonstrated the framework's predictive strength and interpretability.

The LightGBM model, trained on proximity-augmented feature vectors, achieved outstanding performance on the test set, with an RMSE of 0.188, MAE of 0.145, and an R^2 score of 1.000. These results indicate near-perfect alignment between the learned function and the true SoC behavior, affirming both the robustness of the model and the relevance of the engineered features.

SHAP-based feature attribution revealed that `CapaCitymAh` and `VoltageV` are the most influential features. These findings are consistent with the electrochemical characteristics of lithium-ion batteries, where capacity reflects cumulative discharge and voltage tracks internal state. While `CurmA`, `TemperatureC`, and `Cycle` had minimal direct SHAP importance, their effects appear indirectly encoded within the dominant variables, especially through capacity and voltage.

The `Cluster` feature, though not high in SHAP importance rankings, played a crucial structural role. It encapsulated latent regimes, such as low-temperature/high-load conditions, facilitating a more informative feature space. This latent encoding supports the notion that proximity-based partitioning can enhance model learning, even without strong individual predictive power.

The integration of SHAP explanations allowed for transparent evaluation of the model's behavior, a key requirement for critical applications such as electric vehicles or aerospace systems, where prediction trustworthiness is paramount.

In summary, the descriptive clustering combined with interpretable machine learning provides a reliable and transparent SoC estimation approach. Future practical enhancements may involve:

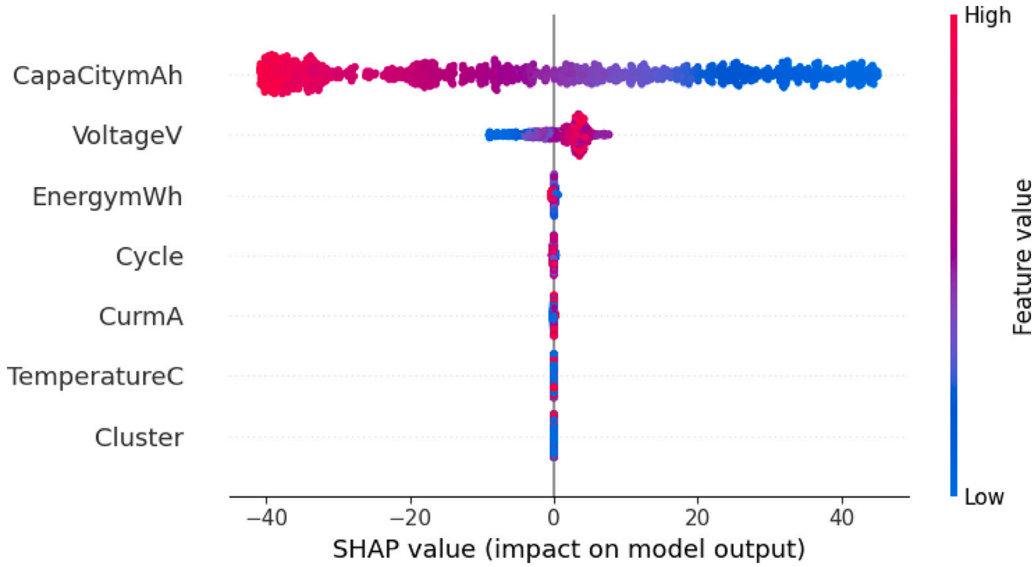

Fig. 6. SHAP summary plot for the SoC prediction model. Top: global feature importance. Bottom: feature value impact distribution.

(*i*) Integrating sequence models (e.g., LSTM) to capture temporal SoC trends.

(*ii*) Adapting the proximity framework to fuzzy or probabilistic clustering for smoother regime modeling.

(*iii*) Extending the validation to broader datasets and different battery chemistries.

In addition to the high baseline accuracy, the robustness analyses provide critical evidence of DPLM's reliability under realistic constraints. The subsampling experiments demonstrated that the model preserves stability even when trained with significantly fewer data, reflecting resilience to limited dataset availability in practical applications. Similarly, the noise injection tests showed that predictive performance remains nearly unaffected by sensor perturbations, which are inevitable in embedded battery management systems. These findings confirm that DPLM is theoretically sound and empirically reliable. Moreover, it remains robust from controlled laboratory tests to real-world applications with noisy data. In this way, robustness verification elevates the framework from a high-performing predictive model to a practically deployable solution for safety-critical energy storage systems.

7.1. SHAP summary visualization

Fig. 7 illustrates the SHAP summary plot generated from the vanilla LightGBM model. Each dot corresponds to a prediction instance, color-coded by the feature value (blue = low, red = high). The horizontal position reflects the SHAP value, representing the impact on the model's output.

Visual Insight. Similar to the DPLM model, `CapaCity(mAh)` and `Voltage(V)` remain the most influential features. However, in the absence of cluster-level structural information, the feature impact appears more concentrated and less distributed across supporting variables such as `Energy(mWh)`, `Cycle`, or `TemperatureC`.

7.2. Quantitative SHAP comparison

To numerically assess the importance of each feature, we computed the mean absolute SHAP value (mean|SHAP|) for both models. The results are summarized in Table 9.

Table 10 compares the SHAP summary plots of the proposed DPLM model and the baseline LightGBM model, highlighting the structural enhancements brought by proximity-based features.

Table 9
Mean absolute SHAP values for DPLM vs. vanilla LightGBM.

Feature	DPLM (Proximity-Enhanced)	Vanilla LightGBM
Capacity (mAh)	**23.82**	21.74
Voltage (V)	**3.56**	2.88
Energy (mWh)	**0.46**	0.31
Cycle	**0.12**	0.09
Current (mA)	**0.07**	0.06
Temperature (°C)	**0.03**	0.02
Cluster (categorical)	**0.21**	–

8. Benchmarking and comparative analysis

To assess the efficacy of the proposed descriptive proximity-enhanced SoC estimation model, referred to hereafter as the **Descriptive Proximity-based LightGBM Model (DPLM)**, we compared its performance against several well-established regression algorithms, including Support Vector Regression (SVR), Random Forest (RF), and a baseline LightGBM model without proximity-based feature augmentation. All models were trained and tested using identical data splits and input features, except for DPLM, which includes an additional cluster label derived from descriptive proximity analysis. The performance metrics used were RMSE, MAE, and R^2. The results are summarized in Table 11.

To better understand the model's impact, a comparison with baseline and classical methods was conducted. The proposed DPLM model significantly outperformed all baseline models, including LightGBM without proximity features. The performance gain over baseline LightGBM-evident from the reduction in RMSE (0.267 to 0.188) and MAE (0.216 to 0.145)-demonstrates the effectiveness of descriptive clustering in enhancing the feature space. Compared to SVR and RF, which are classical regression methods, DPLM delivers superior accuracy and consistency across all metrics.

In particular, the improvement from baseline LightGBM to DPLM highlights the added value of incorporating descriptive cluster labels as structural features. These clusters capture underlying variations in discharge behavior caused by thermal and electrical conditions, which are otherwise difficult to model using raw features alone.

Beyond predictive accuracy, computational efficiency is also an important consideration. It is worth noting that LightGBM-based models (both the baseline and DPLM) surpass SVR and RF in terms of both

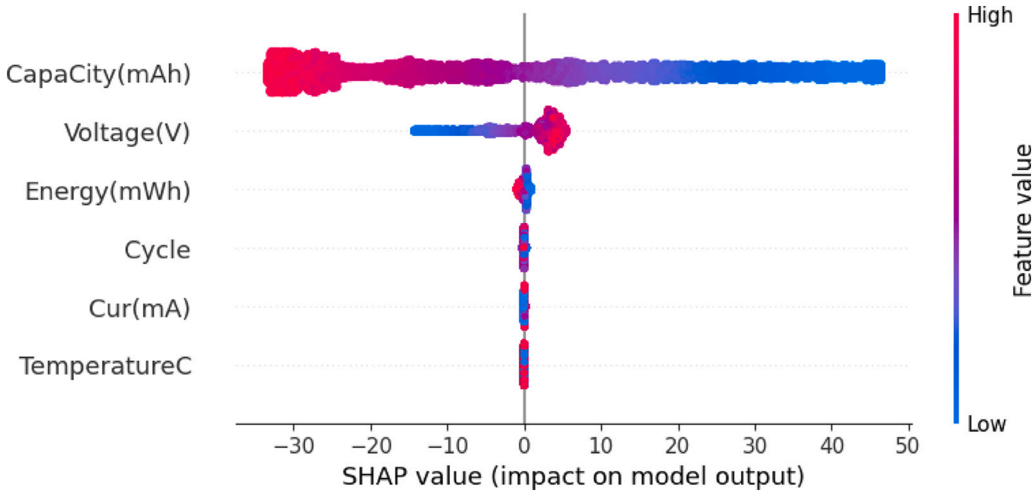

Fig. 7. SHAP summary plot for the vanilla LightGBM model without proximity-based enhancements. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 10
Comparison of SHAP Explanations: DPLM vs. Vanilla LightGBM.

Feature	Fig. 6 (DPLM Model)	Fig. 7 (Vanilla LightGBM)
SHAP distribution	More **widespread**, distributed across multiple relevant features	More **concentrated**, dominated by a few primary features
Cluster effect	Present (final feature in list: Cluster)	Absent (no proximity-based structural features included)
Role of Capacity & Voltage	Still dominant, but **other features also gain explanatory strength**	Capacity overwhelmingly dominant, **Voltage less influential**
SHAP value spread	Broad range, including outliers in both positive and negative space	Narrower distribution, centered closer to zero
Interpretive message	Proximity + clustering **enrich feature attribution and diversity**	Without structural context, **only core features drive predictions**

Table 11
Comparison of SoC prediction models trained on the same dataset. DPLM integrates descriptive proximity features.

Model	RMSE	MAE	R^2
DPLM (Proposed)	**0.188**	**0.145**	**1.000**
LightGBM (Baseline)	0.267	0.216	0.998
Random Forest	1.052	0.842	0.962
Support Vector Regression (SVR)	1.363	0.992	0.942

accuracy and computational speed. This advantage makes DPLM particularly well-suited for deployment in real-time battery management systems.

Taken together, these benchmarking results provide strong empirical support for the integration of descriptive proximity theory into modern machine learning pipelines. The DPLM model not only advances predictive accuracy but also enhances model transparency and contextual adaptability in SoC estimation.

9. Concluding remarks and theoretical implications

This study proposed the DPLM, an integrated framework combining adaptive unsupervised clustering, fuzzy relational reasoning, and gradient-boosted regression for accurate and explainable SoC estimation in LIBs. The model maps battery discharge data into a descriptive feature space $\Phi(x) = (\text{current}, \text{voltage}, \text{capacity}, \text{energy}, \text{cycle count}, \text{temperature})$, within which local proximity reflects similar operational regimes.

A core theoretical innovation lies in the use of adaptive DBSCAN clustering, driven by a data-sensitive threshold $\varepsilon_{\text{adaptive}} = 1.4826 \times$ MAD, eliminating the need for manual parameter tuning and enabling meaningful cluster formation that corresponds to temperature- and current-induced variations. To evaluate the cohesion of these clusters, we defined a fuzzy proximity relation $\mu_R(A, B)$, quantifying similarity between point pairs through an exponential formulation. This allowed for rigorous verification of intra-cluster compactness and inter-cluster separability, reinforcing the structural validity of the resulting regimes.

Although the fuzzy proximity metric was not directly integrated into the LightGBM training process, its analytical value is twofold: it validates clustering coherence and offers future potential for soft cluster membership encoding, probabilistic regime modeling, and domain adaptation across diverse battery chemistries and degradation levels. Embedding fuzzy proximity into the learning process-either as feature augmentations or through relational regularization-could enhance resolution in transitional or ambiguous SoC states, bridging the gap between quantitative prediction and topological semantics.

Empirical results show that DPLM achieves state-of-the-art predictive performance, with RMSE $\approx$ 0.188, MAE $\approx$ 0.145, and $R^2 \approx$ 1.00, remaining consistent across 5-fold cross-validation. Comparative benchmarks confirm its superiority over traditional models (Random Forest, SVR), baseline LightGBM, and even recent deep learning alternatives such as LSTM-RNN and GRU-RNN, while offering significantly improved model interpretability. In particular, SHAP attributed SoC prediction to physically meaningful features such as capacity and voltage, aligning model behavior with electrochemical principles.

The structural design of DPLM also supports operational efficiency. Its low complexity and compact memory footprint make it well suited for deployment in embedded systems, including electric vehicles, drones, and smart grid applications. Additionally, descriptive clustering serves not only to enhance interpretability but also opens pathways for predictive maintenance by identifying stress-related degradation regimes.

Furthermore, the framework demonstrated strong robustness in additional validation tests. Subsampling experiments confirmed that predictive accuracy remains stable even with reduced data availability, while noise robustness analysis showed resilience to sensor perturbations with R^2 consistently above 0.9997. These findings reinforce the generalizability and reliability of DPLM under realistic operating scenarios.

Despite these strengths, the framework has certain limitations. It has been validated primarily on cylindrical 18 650-type Li-ion cells under limited thermal and electrical conditions. Broader generalization across chemistries such as LFP, NCA, or LCO, and under varying stress conditions, remains a key future goal. Furthermore, while the current model effectively captures static nonlinearities, it lacks temporal modeling capability, which is critical for capturing degradation dynamics in long-term operation.

Looking ahead, several promising extensions are envisioned:

- Integration of temporal models (e.g., LSTM, Bi-LSTM) to capture sequential patterns in degradation behavior.
- Expansion to multi-cell configurations and variable battery topologies to support large-scale energy storage systems.
- Incorporation of fuzzy proximity metrics as soft constraints or features to enable nuanced regime transitions and uncertainty quantification.
- Adaptation to online and streaming learning settings for real-time monitoring and decision-making in safety-critical environments.

In conclusion, the DPLM framework presents a theoretically grounded, practically viable, and computationally efficient solution for interpretable SoC estimation. Its fusion of symbolic reasoning and machine learning offers a new paradigm for battery analytics, paving the way for intelligent, structure-aware energy storage management systems.

CRediT authorship contribution statement

Ebubekir İnan: Writing – original draft, Methodology, Formal analysis, Conceptualization. **Ahmet Aksöz:** Writing – review & editing, Validation, Resources, Investigation, Data curation, Conceptualization. **Saadin Oyucu:** Writing – review & editing, Visualization, Validation, Software. **Emre Biçer:** Writing – review & editing, Validation, Investigation, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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